QUALITY INFORMATION DOCUMENT

For
OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_sto2tz_gotland

Issue: 4.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/06	all	Creation of the document	U. Raudsepp I. Maljutenko J. Linders	P. Lagemaa
1.1	2019/02/21		Revision	U. Raudsepp I. Maljutenko	P. Lagemaa
2.0	2019/10/10		Update to 2018 data	U. Raudsepp I. Maljutenko	P. Lagemaa
2.1	2021/10/01	all	Remove temporal extension specific details	U. Raudsepp I. Maljutenko	P. Lagemaa
3.0	2023/09/01	all	Template change+ minor changes	U. Raudsepp I. Maljutenko	P. Lagemaa
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1) INTRODUCTION

a) Product Name

OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_sto2tz_gotland.

b) Indicator Evaluation Method

This document provides to Copernicus Marine Service users information on the quality of the time-depth plots of salinity, temperature and dissolved oxygen in the Gotland Basin. These quantities were chosen to represent the inflow events of the Baltic Sea and therefore grouped as a separate Copernicus Marine Service product which is named BALTIC_OMI_WMHE_mbi_sto2tz_gotland.

The time-depth data of the salinity, temperature and dissolved oxygen are extracted from INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032 product.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032 (insitu observations)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-INS-QUID-013-030-036.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-INS-PUM-013-030-036.pdf

2) SCIENTIFIC VALIDATION

a) Baltic Sea Major Baltic Inflow: time/depth evolution S,T,O2

The accuracy of the index is directly dependent on the accuracy of observation data (product INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032). The index is constructed directly from CTD data at the Gotland deep monitoring station. According to the QUID of INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032 the accuracy of CTD temperature, salinity and oxygen observations are 0.005 – 0.001 ˚C, 0.02 – 0.003 and 8 mMol, respectively (Figure 1).

Scientific evaluation can be found in the Copernicus Marine Service Ocean State Report (Raudsepp et al., 2016; Raudsepp et al., 2018).

The observation data from monitoring station at the Gotland deep (in the Gotland Basin) have been quality checked and valid therefore. The sampling is performed from a research vessel with a CTD Sbe911+, the temperature sensor is a Sbe3+, the conductivity sensor is a Sbe4C and the oxygen sensor is a Sbe43. The CTD and associated sensors are served align with the recommendations of the manufacturer.

Directly after a CTD cast is taken the data is automatically processed via a scheme within the SbeDataProcessing tool, thereafter the profiles for all parameters are visually inspected to sort out instabilities, spikes, frozen profiles, invalid ranges and to check the difference between the two sensor packages for salinity, temperature and oxygen. The observations are then sent as quickly as possible, depending on internet connection, to our ftp to be disseminated in near real time.

The profiles are then later on, in normal within three months from measure date, manually checked once again and data are compared with bottle data and thermometer data from external thermometers situated on the bottles. The profiles are stored at an internal file server, and disseminated via Copernicus Marine Services.

Table 2: Accuracy numbers for temperature and salinity observations in the different platforms of the In Situ TAC.

Data-type	Temperature[°C]	Salinity []
CTD (Conductivity Temperature Depth)	0.005 - 0.001	0.02-0.003

Table 5: Accuracy numbers for biogeochemical parameter observations in the different platforms of the In Situ TAC.

Data-type	**Chlorophyll fluorescence** Exemplary TriOS Microflu sensor. There is quite a range of sensors with quite a variety of accuracy available.	**Oxygen** Values given by a number of sensor providers, Aanderaa Instruments, Endress and Hauser.	**Nutrients[µmol kg^{-1}]** Exemplary Systea MicroMac sensor. There is a limited number of different sensors available.
CTD (Conductivity Temperature Depth)	0.05	8 myMol/5% 0.2 %	Discrete samples

Figure 1. The accuracy numbers of CTD data. Reprint from QUID of INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032.

3) REFERENCES

Raudsepp et al., 2016: Major Baltic inflows. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 1, Journal of Operational Oceanography, Volume 9, 2016 - Issue sup2: The Copernicus Marine Environment Monitoring Service Ocean, http://dx.doi.org/10.1080/1755876X.2016.1273446.

Raudsepp, U., Legeais, J.-F., She, J., Maljutenko, I., Jandt, S., 2018. Baltic inflows. In: Copernicus Marine Service Ocean State Report, Issue 2, Journal of Operational Oceanography, 11:sup1, s106–s110, DOI: 10.1080/1755876X.2018.1489208